



GIEWS Update

Agricultural production risks associated with El Niño conditions and shipping disruptions linked to the Middle East conflict

Highlights

- The ongoing El Niño event is expected to persist through at least early 2027, bringing an increased risk of extreme weather events that could adversely affect agricultural production in several regions.
- Dry weather conditions are forecast in parts of sub-Saharan Africa, Far East Asia and Central America, while above-average rainfall amounts are foreseen in Near East Asia.
- The impacts of the shipping disruptions in the Strait of Hormuz could exacerbate El Niño-related production risks, through shortages and high prices of agricultural inputs. While the United States of America and the Islamic Republic of Iran signed a Memorandum of Understanding on 17 June 2026, the economic, trade and geopolitical repercussions of the conflict are expected to last longer.

Overview

The emergence of El Niño conditions in 2026 is expected to increase the probability of weather anomalies across several major agricultural producing regions worldwide, with potential implications for crop and livestock production. Below-average rainfall amounts in some areas may constrain yields, reduce plantings and affect pasture conditions, while above-average rainfall amounts elsewhere could boost yields but may raise the risk of flooding and localized production shortfalls.

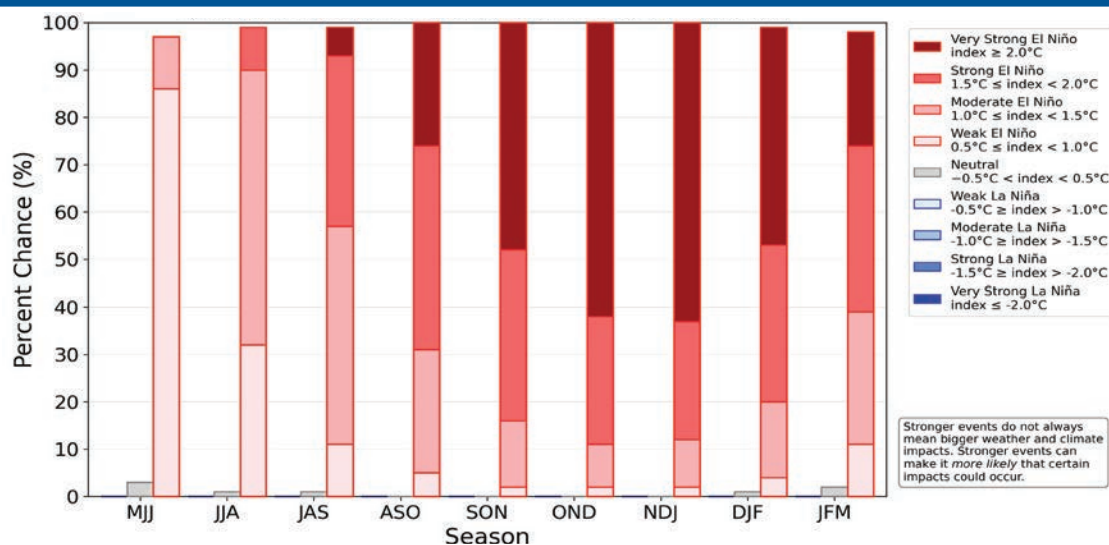
At the same time, disruptions in maritime shipping through the Strait of Hormuz since late February 2026 have affected global energy and fertilizer markets, and contributed to higher agricultural input and transport costs. Reduced availability and higher prices of fuel and fertilizers during key agricultural periods may limit input use and affect production outcomes, particularly in import-dependent countries. Whilst the recently announced preliminary peace agreement in mid-June 2026, which foresees the reopening of the Strait of Hormuz upon formal signing, is a positive development, the full resumption of traffic flows is expected to only normalize progressively.

This update provides an overview of countries expected to be most exposed to El Niño-induced weather anomalies and disruptions to trade flows through the Strait of Hormuz, with particular attention to areas where dry weather conditions may coincide with constrained access to agricultural inputs.

El Niño impact

According to the latest forecasts as of June 2026, El Niño conditions are present and are expected to persist until at least the January-to-March 2027 period. (Figure 1).ⁱ Overall, El Niño is typically associated with below-average rainfall amounts in parts of West Africa, central and northern parts of East Africa, Southern Africa, western areas of East Africa, India, Southeast Asia, Australia, Central America and northern areas of South America. Conversely, above-average precipitation amounts and associated risks of flooding are generally observed in Central Asia, the Near East, southern parts of South America and the United States of America, and eastern areas of East Africa (Map 1). At the same time, climate models indicate the likely development of a positive Indian Ocean Dipole (IOD) during June to August 2026 period, with the event expected to strengthen alongside the developing El Niño.ⁱⁱ This combination

Figure 1. ENSO probability forecast, June 2026



Note: NOAA/CPC ENSO Forecast Graphic, courtesy of NOAA/CPC. Based on thresholds in ERSSTv5 Relative Niño-3.4 index/RONI.

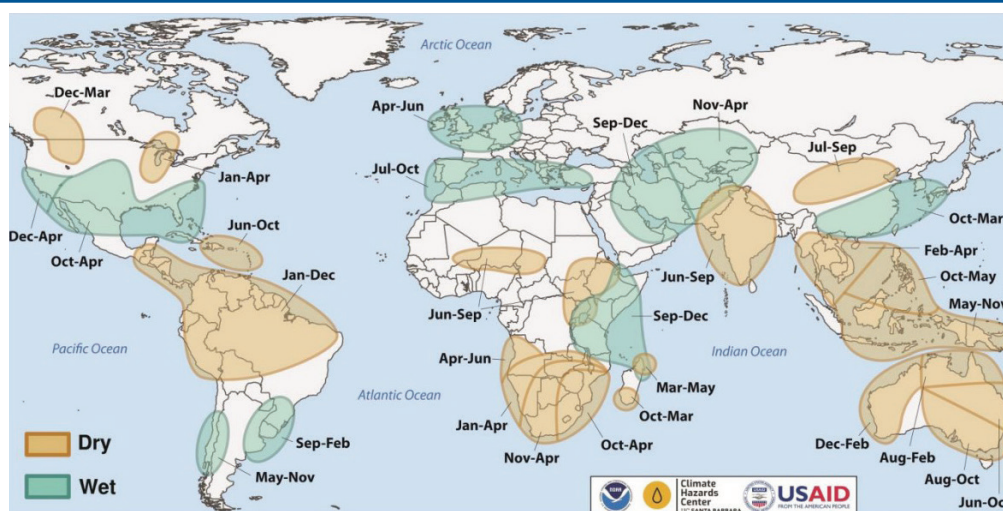
Source: CPC. 2026. *Official NOAA CPC ENSO Strength Probabilities*. NOAA's Climate Program Office and Columbia University. International Research Institute for Climate and Society (IRI). June 2026. https://cpc.ncep.noaa.gov/products/analysis_monitoring/ens0/roni/strengths/

usually amplifies rainfall deficits across parts of Asia and Australia, and increases wet conditions in eastern parts of East Africa.

Areas experiencing severe dry weather conditions are of particular concern, given the multiple risks associated with water stress, including reduced crop yields, lower planting intensity, diminished pasture and livestock productivity, and increased risk of crop failure. Of particular concern, drier-than-average conditions are expected to impact entire cropping

cycles in some countries located in Central America, southern areas of Far East Asia, Southern Africa, central and northern parts of East Africa, and northern parts of South America. By contrast, above-average precipitation amounts are expected in eastern parts of East Africa, Central Asia, the Near East, southern parts of South America and the United States of America, with generally favourable implications for crop and livestock production. However, excessive rainfall may also increase the risk of flooding and waterlogging, potentially causing localized crop losses and negatively

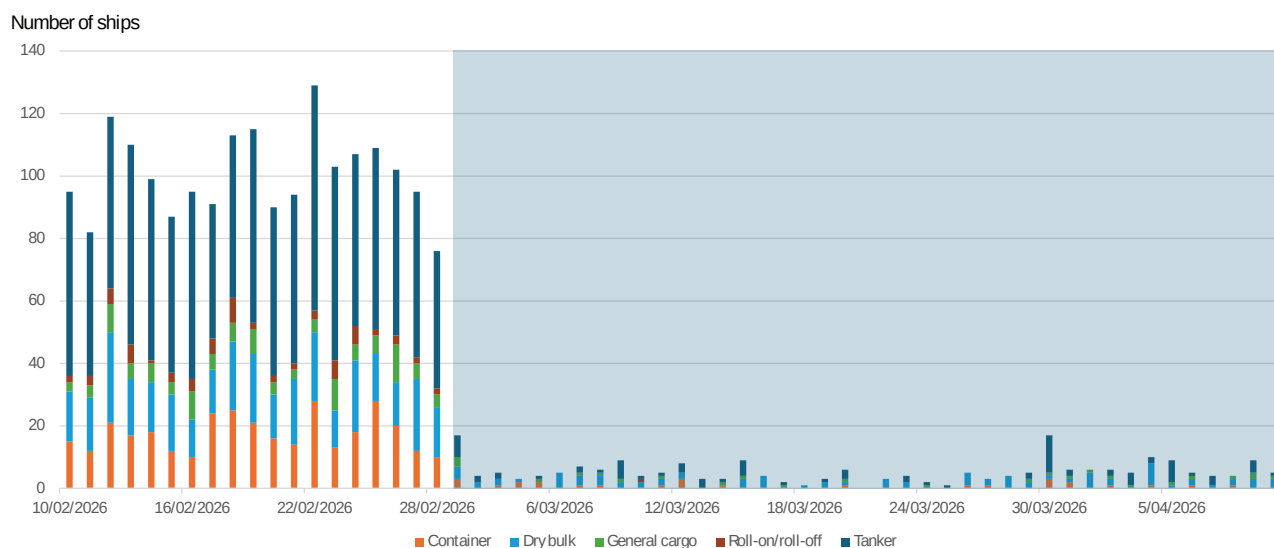
Map 1. Locations and seasonal patterns of precipitation anomalies associated with El Niño



Notes: Refer to the disclaimer on [back cover](#) for the names and boundaries used in this map. Final boundary between the Republic of the Sudan and the Republic of South Sudan has not yet been determined. Dotted line represents approximately the Line of Control in Jammu and Kashmir agreed upon by India and Pakistan. The final status of Jammu and Kashmir has not yet been agreed upon by the parties

Source: FEWS NET. 2026. *Famine Early Warning Systems Network, 2020: El Niño and Precipitation, Agroclimatology Fact Sheet Series, Vol. 1, pp. 1*. Washington, D.C. August 2026. <https://fews.net/sites/default/files/2025-08/White%20Paper%20-%20El%20Nino.pdf>

Figure 2. Ship traffic through the Strait of Hormuz, 2026



Source: Author's own elaboration based on data from the IMF. 2026. *IMF Port Watch: Port Monitor Strait of Hormuz*. Washington, D.C. [Cited May 2026]. <https://portwatch.imf.org/pages/chokepoint6>

impacting livestock production in the affected areas, as excessive soil moisture creates favourable conditions for the spread of crop pests, weed proliferation as well as crop and animal diseases.

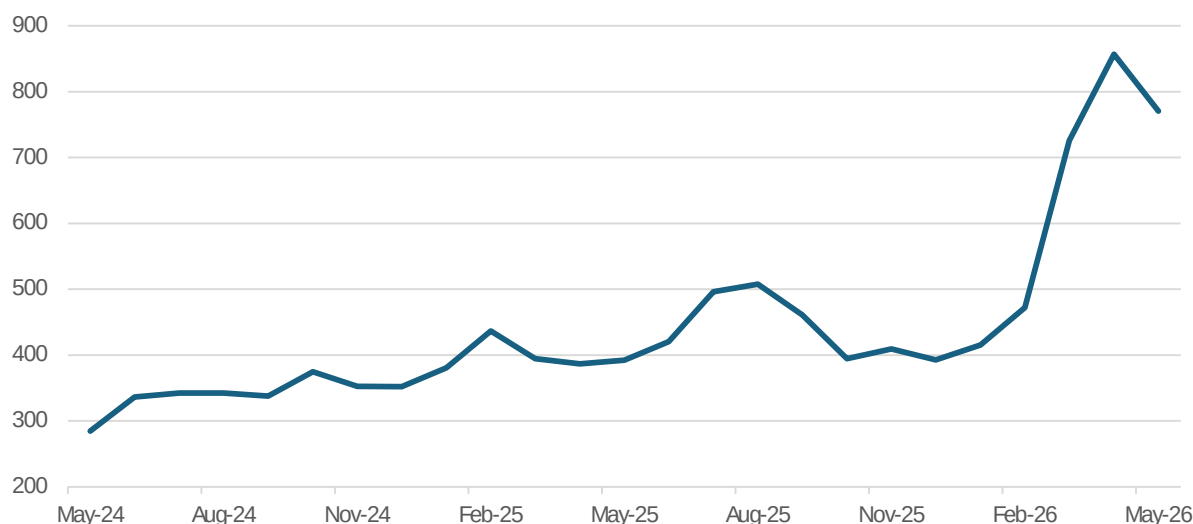
Shipping restrictions in the Strait of Hormuz

The Strait of Hormuz is a critical route and hub for global energy and commodity trade, with nearly one-quarter of the world's seaborne oil shipments, alongside substantial volumes of liquefied natural

gas (LNG) and fertilizers.ⁱⁱⁱ In 2024, about 30 percent of global fertilizer trade from the Persian Gulf passed through the strait.^{iv} The recent conflict in the Middle East disrupted shipping flows along this corridor. These disruptions have had extended global repercussions, affecting energy markets, maritime logistics and global supply chains for energy and fertilizers, contributing to a surge in agricultural input prices.

Global crude oil prices were up by more than 50 percent in May compared to February 2026 (Figure 3). Following the outbreak of conflict,

Figure 3. Average international crude oil prices, May 2024–May 2026



Source: Author's own elaboration based on data from the World Bank. 2026. *Commodity Markets: World Bank Commodity Price Data (The Pink Sheet)*. World Bank Group. Washington, D.C. [Cited May 2026]. <https://www.worldbank.org/en/research/commodity-markets>

Figure 4. International urea export prices (Black Sea), February 2024–May 2026



Source: Author's own elaboration based on data from the FAO. 2026. *Food Price Monitoring and Analysis (FPMA) Tool: International Prices*. Rome. [Cited May 2026]. <https://fpma.fao.org/gIEWS/fpmat4/global/#/dashboard/tool/international>

significantly increasing cost pressures on food supply chains, from agricultural operations to transport and storage, and thereby raising the risk of renewed food price inflation. Oil markets remain highly volatile, amid concerns over potential continued supply disruptions in the Persian Gulf.

The disruptions to LNG production and exports from the Persian Gulf, notably from Qatar, have exerted significant pressures on global fertilizer supply chains and constrained nitrogen production. Countries that rely heavily on gas supplies from the region for fertilizer manufacturing, have faced significant challenges in meeting their domestic energy needs and sustaining fertilizer production.^v

Beyond energy, the Persian Gulf is a key hub for the global fertilizer market. Major exporters, including the Islamic Republic of Iran, Qatar, Saudi Arabia and Oman, accounted in recent years for about 30–35 percent of global urea exports and 20–30 percent of ammonia exports, with most shipments routed through the Strait of Hormuz.^{vi} Disruptions to shipping and production since late February 2026, tightened global supply, contributing to a surge of about 60 percent in international urea prices in May relative to February (Figure 4). This is expected to raise agricultural

production costs in the upcoming season and may, in turn, affect production levels.

Countries that rely heavily on imports to meet their domestic energy and fertilizer requirements are particularly exposed to market disruptions and rising input costs. Several countries depend significantly on fertilizer imports sourced from the Persian Gulf, especially from Saudi Arabia, the Islamic Republic Iran and Qatar. The surge in prices will likely further constrain the already inadequate fertilizer use among smallholder farmers, with immediate implications for agricultural productivity. Reduced fertilizer applications could result in lower yields and domestic production, depending on existing nutrient application rates, crop type and availability of alternative crops. Global food commodity supply could also be affected, especially if production declines in major exporting countries. At the same time, higher energy prices have driven up transport costs, increasing the cost of food imports and adding further pressure on domestic food prices.^{vii} The exact magnitude of these impacts remains uncertain, and whilst the signing of a peace agreement is a welcome development, the full reopening of the Strait of Hormuz and the rebuilding of energy production capacity in the region are expected to only normalize progressively.

Regional impacts

North Africa

In North Africa, El Niño events are generally associated with an increased likelihood of above-average rainfall amounts between July and October, a period that largely falls outside the main cropping season in the subregion. However, countries are vulnerable to disruptions in global food and energy markets due to their strong reliance on cereal imports and exposure to rising fuel prices. Higher international energy prices and market volatility are likely to increase agricultural production costs, while higher food import bills may exert additional domestic inflationary pressure, with adverse impacts particularly on vulnerable households. The degree of exposure, however, varies considerably across countries, depending on the cereal import dependency, energy trade balance, domestic fertilizer production capacity and fiscal space. Morocco, Egypt and Algeria benefit from domestic fertilizer production capacity that may help mitigate the impact of higher international fertilizer prices. Countries with greater reliance on fuel imports, including Egypt, Morocco and Tunisia, are likely to face higher import bills and inflationary pressures when oil prices rise, while energy exporters, notably Algeria and Libya, may benefit from increased export revenues. Overall, countries across the subregion remain vulnerable to external food and energy market shocks that could contribute to rising food prices.^{viii, ix}

West Africa

In West Africa, the impact of El Niño on agroclimatic conditions varies across the subregion and is difficult to predict. In some parts of the Sahelian Belt, El Niño conditions are associated with a higher likelihood of below-average precipitation amounts during the June–September rainy season, which could adversely affect planting activities, reduce cultivated area and lower crop yields. While countries exposed to El Niño-related dry weather risks generally have limited dependence on oil and fertilizer imports transiting through the Strait of Hormuz, high international prices for these inputs are likely to constrain imports and drive up domestic fuel and fertilizer prices. Across the subregion, fertilizer application typically takes place between May and September, meaning that these expected price increases coincide with planting activities and peak input use, and could negatively affect 2026 cereal production prospects.

Among the countries most exposed to potential El Niño-related dry weather conditions are the Niger, where all major producing areas are at risk, and Chad, where western and central areas face an increased likelihood of below-average precipitation amounts. In both countries, millet and sorghum account for the bulk of cereal production and are mainly cultivated under rainfed, low-input systems. Therefore, precipitation deficits are likely to constitute the main risk to production, while the impact of high fuel and fertilizer prices on aggregate cereal production in 2026 is likely to remain limited. By contrast, input-intensive maize and rice crops may be adversely affected by reduced fertilizer use due to high prices and production could decline. In Chad, elevated international oil prices are likely to exert additional upward pressure on the already high domestic fuel prices, further increasing transport and production costs.

In Burkina Faso and Mali, risks to 2026 cereal production are likely to be driven mainly by fertilizer constraints rather than El Niño-related dry weather conditions, which typically affect only localized areas of central and eastern Mali, and northern Burkina Faso. According to reports published in April 2026, available mineral fertilizer stocks for the upcoming agricultural season covered just about 17 percent of the requirements for direct government support to producers in Burkina Faso, while in Mali, stocks covered just 22 percent of the estimated national requirements as of end-March 2026.^{x, xi} High international prices are expected to constrain imports and maintain domestic fertilizer prices at high levels, likely reducing fertilizer application rates and potentially impacting maize profitability and planting decision in both countries, while also affecting irrigated cereal production in Mali's Niger River Basin, the country's main rice-producing area. In addition, in Mali, high international oil prices have exacerbated the already strained domestic fuel situation, further increasing transport and irrigation costs.

In Nigeria, El Niño-related dry weather conditions are typically confined to localized areas in the far north and are, therefore, unlikely to significantly affect cereal production in 2026. However, despite being a major crude oil producer, the country remains exposed to international energy price shocks due to its reliance on refined fuel imports. Recent increases in global energy prices have already contributed to the sharp rises in domestic fuel prices, which in turn have raised transport

costs and fertilizer prices. Although the significant domestic urea production and fertilizer blending capacity are expected to support fertilizer availability, persistently elevated energy and logistics costs are likely to keep fertilizer prices high, thereby constraining input use during the 2026 agricultural season.

Central Africa

In Central Africa, risks to cereal production are driven mainly by climatic exposure due to the El Niño conditions, as the impact of fertilizer supply disruptions remains limited, reflecting the predominance of low input agricultural systems and a low dependence on imports transiting through the Strait of Hormuz. Nevertheless, this does not reduce the vulnerability of countries to external shocks. Across large parts of the Democratic Republic of the Congo below-average rainfall amounts are forecast during the June–October period, which coincides with the planting season of the 2026 main season maize crop, accounting for about 80 percent of the annual production, as well as the sowing of sorghum and millet crops, while current forecasts do not point to similar weather-related concerns in the Central African Republic, Cameroon, Gabon and Equatorial Guinea.^{xii}

East Africa

El Niño conditions are typically associated with uneven rainfall patterns across East Africa, with impacts varying according to the timing of the rainy seasons and crop calendars. Consistent with the June–September 2026 seasonal forecast issued by the Intergovernmental Authority on Development's (IGAD) Climate Prediction and Application Centre (ICPAC), below-average rainfall amounts are likely to affect crops in key cereal-producing areas dependent on the June–September rains, including southeastern Sudan, western Ethiopia, South Sudan and Eritrea. By contrast, El Niño conditions are generally associated with above-average rainfall amounts between September and December over most of Somalia, eastern and northern Kenya, southern and eastern Ethiopia as well as Uganda, the United Republic of Tanzania, Rwanda and Burundi.

Risks to cereal production in the Sudan and Ethiopia reflect both climatic shocks and disruptions to fertilizer availability and prices, as about one-quarter and one-fifth, respectively, of their fertilizer imports transit through the Strait of Hormuz.^{xiii} In the Sudan, these risks are concentrated in key southeastern

producing areas, where a large share of the commercially-oriented cereal production is located. In Ethiopia, there are concerns for the most important cereal-producing areas, including western Oromia, western Amhara and Benishangul Gumuz regions, which depend on the June–September *Kiremt* rains for the cultivation of *Meher* crops. The presence of El Niño conditions from June onwards is, therefore, likely to coincide with a critical stage of the growing season in both countries, increasing the risk of below-average cereal production. Concerns over fertilizer supply disruptions and adverse climatic conditions may also extend to mechanized production areas in Upper Nile State of South Sudan (Renk and Melut counties), which depend on inputs sourced from the Sudan and are largely oriented towards Sudanese markets.

In Kenya, risks to cereal production arise primarily from input-related constraints and, to a lesser extent, climatic factors. Almost 40 percent of the country's fertilizer input imports transit through the Strait of Hormuz, particularly affecting market-oriented production areas of central, Rift Valley and western regions, where the main *long-rains* season typically extends from February to August. Delays in fertilizer imports have already disrupted timely application at the start of the 2026 cropping season and are likely to weigh on yields.^{xiv} The presence of El Niño conditions are expected to affect only the latter part of the cropping cycle, thereby limiting the overall impact of climatic factors.

By contrast, in South Sudan and Eritrea, risks to cereal production are driven mainly by climatic exposure due to the El Niño conditions, as the impact of fertilizer supply disruptions remains limited, reflecting the predominance of low input agricultural systems. In South Sudan, below-average rainfall amounts between June and September are expected to affect crops in central and northern *unimodal* rainfall areas, where the main growing season typically spans from June to October. In southern *bimodal* areas, the first season (May–July) is likely to be only partially affected, while the second season (October–December) is expected to be affected throughout its cycle. In Eritrea, below-average rainfall amounts during the June–September *Kiremti* rainy season are expected to adversely affect cereal production.

In the remaining countries of the subregion, including the United Republic of Tanzania, Somalia, Uganda, Rwanda and Burundi, expected above-average

rainfall amounts between September and December are generally expected to support yields of the 2026 second/secondary season crops and the 2027A season crops in Rwanda and Burundi. However, in the United Republic of Tanzania and Uganda, where about one-third of fertilizer imports transit through the Strait of Hormuz,^{xv} the favourable weather effects on yields may be partially offset by reduced fertilizer application due to high prices and supply disruptions.

Across the subregion, rising international fuel prices are increasing production and transport costs, and are expected to continue exerting upward pressure on domestic food prices, which are already at high levels.

Southern Africa

In Southern Africa, the primary risk to crop production is weather-related. Elevated fertilizer prices and tightening availability constitute a secondary but compounding factor, with their impact tempered by the predominantly low-input nature of agricultural systems across the region. As harvesting of the 2026 main season cereal crops is expected to conclude in June, these risk factors are expected to primarily influence production of 2027 cereal crops, to be planted from October 2026 onwards. Maize, the main food staple within the subregion, is a particularly vulnerable crop, given its relatively high susceptibility to water stress and comparatively high input requirements.

El Niño is typically associated with below-average rainfall amounts throughout the main cropping season, which runs from October to April. This rainfall pattern affects most parts of the subregion, although in Angola and Madagascar the impacts tend to be more limited. Water deficits during the 2026/27 cropping season could, therefore, curtail both plantings and yields, resulting in lower cereal production across most of the subregion. The case of South Africa is particularly important, given its role as a source of exports for neighbouring countries, and production shortfalls there would have both domestic and subregional implications. The scope to offset rainfall deficits is limited, in consideration of the relatively small share of irrigated cropland. Irrigation coverage is comparatively higher in South Africa and Eswatini than in most other countries in subregion; however, past El Niño episodes have resulted in steep cereal production downturns, notably in 2016 and 2024, suggesting that mitigation capacity remains constrained.

Regarding rising input costs, South Africa's cereal production is the most exposed to high fertilizer costs and tightening supply. The country has the highest fertilizer application rate in Southern Africa and nearly one-third of its inorganic fertilizer supplies come through the Strait of Hormuz.^{xvi} Continued high input costs, with local urea prices already about 90 percent higher year-on-year in May,^{xvii} combined with expected lower grain prices in 2026 due to a record harvest and large carryover stocks, would likely compress profit margins for the 2027 maize crop. This may curb area expansion and prompt farmers to reduce fertilizer application rates. Although lower fertilizer use could weigh on yields, the extent of the impact is uncertain. This is because yield responses to nitrogen fertilizer (a key input in maize production) are generally weaker at higher application rates, as in the case of South Africa, therefore, modest reductions from the already high fertilizer use levels are unlikely to have a significant effects on yields.^{xviii} Impacts from higher fertilizer prices in the remaining countries of the subregion are expected to be more limited and more heterogenous than weather-related risks. Farmers in Malawi and Zambia are, however, still expected to face notable exposure to rising fertilizer costs due to their relatively high application rates by regional standards. Government subsidy programmes may partly buffer the increased costs, but potential fertiliser supply constraints, as well as reduced affordability, may adversely impact planting decisions. In most other countries in the subregion, overall fertilizer use is limited, reducing direct exposure to higher fertilizer prices. However, lower fertilizer use also reflects broader structural constraints, including limited input access and lower yield potential. These constraints, in general, heighten farmers' vulnerabilities to adverse weather conditions, as they limit capacity to adopt risk mitigating strategies.

Although tentative and preliminary, early production prospects suggest an increased risk of below-average production in 2027. Such declines could worsen acute food insecurity by reducing food availability and constraining access, through lower on-farm incomes and potentially rising food prices. However, current ample subregional cereal supplies and the expected bumper harvests in 2026, including likely records in South Africa and Zambia, are likely to help cushion these effects and in part stabilize supplies in 2027.

Far East Asia

In Far East Asia, El Niño-induced below-average precipitation amounts are expected in several parts of the subregion from June to September 2026, coinciding with the 2026 main season in the Northern Hemisphere countries and may constrain plating activities and reduce yield potential for rice and coarse grains. Below-average precipitation amounts are expected in parts of India, Bangladesh, Bhutan, Cambodia, Lao People's Democratic Republic, Malaysia, Nepal, the Philippines, Thailand, eastern Pakistan, central and southern Myanmar, and southern Viet Nam. In countries south of the Equator, including Indonesia and Timor-Leste, below-average rainfall amounts are likely from June 2026 to January 2027, likely affecting the 2026 secondary crops from vegetative stage to harvest, while also constraining planting operations of the 2027 main crops, which typically begin in October. By contrast, above-average precipitation amounts are expected across most of Afghanistan and northern Pakistan, with potential benefits for 2027 wheat crops.

Several countries that are expected to face El Niño-induced unfavourable weather conditions rely heavily on fertilizer imports via the Strait of Hormuz, notably Afghanistan (about 55 percent), Bangladesh (about 40 percent) and India, Thailand, Sri Lanka and Pakistan (about 30 percent).^{xix} Within the subregion, Afghanistan, Bangladesh and Myanmar are especially exposed to the combined effects of El Niño-induced weather anomalies and disruptions to critical inputs. In Bangladesh, about 80 percent of fertilizer requirements are imported, with about half of it transiting through the Strait of Hormuz.^{xx, xxi}

In Afghanistan, the heavy reliance on imported food, fuel and fertilizer means that disruptions to their supply risk worsening already elevated levels of acute food insecurity. In Myanmar, macroeconomic challenges could limit the country's capacity to secure sufficient fuel and agricultural inputs. In Pakistan, urea requirements are largely met through domestic production, while phosphate and potash fertilizers, important inputs for major crops such as wheat and rice, depend largely on imported products and raw materials. Fertilizer stocks are currently sufficient for the 2026 *Kharif* season, which extends from June to November.^{xxii} However, disruptions to regional trade flows and elevated energy prices could increase

fertilizer costs in the second half of 2026, potentially affecting input affordability for the 2027 wheat crop, to be planted from October onwards.

While the 2026 subregional wheat production is forecast at above-average levels as crops were already at harvesting stages when Hormuz-related disruptions occurred, the production outlook for 2026 rice and maize crops is more uncertain. Their main and secondary production seasons extend from June 2026 to early 2027, coinciding with the negative effects of El Niño and potential disruptions to fuel and fertilizer shipments. Potential production shortfalls in major exporting countries of the subregion, such as India, Thailand and Viet Nam, could further push up international cereal prices, which have already increased since February 2026, raising food import bills for import-dependent countries. However, ample subregional stocks following the record cereal production in 2025 are expected to partially buffer these effects in the near term.

Near East and Central Asia

In the Near East and Central Asia, countries in the Gulf Cooperation Council (GCC), Saudi Arabia, the United Arab Emirates, Qatar, Kuwait, Bahrain and Oman, are highly dependent on food imports (about 70–90 percent), reflecting limited domestic agricultural production due to scarce arable land, water shortages and climatic factors. Despite their high income status, disruptions to maritime trade routes through the Strait of Hormuz threaten their primary food supply channel. Although GCC countries maintain strategic food reserves for four to six months of consumption, a prolonged disruption could strain supply chains and contribute to rising domestic food prices as these reserves are gradually depleted.^{xxiii}

In the short-term, GCC countries are implementing measures to sustain supply flows, including rerouting imports through alternative ports, including Oman, and expanding the use of air freight for essential goods. While these measures help address immediate constraints, they are costly and cannot fully substitute for regular import volumes through Persian Gulf ports. If disruptions persist, supply chains could tighten and prices rise further. Overall, this situation underscores the structural reliance of the subregion on food imports despite its short-term resilience.

Elsewhere in the subregion, including Lebanon, the Syrian Arab Republic and Yemen, limited domestic production, inadequate storage capacity and persistent economic crises heighten exposure to high international food prices. Rising energy prices and transport costs are expected to further increase import bills and place additional upward pressure on domestic food prices. At the same time, currency volatility and constrained foreign exchange availability are compounding these effects, while ongoing conflicts and economic challenges continue to disrupt livelihoods and erode household purchasing power. Together, these factors risk worsening acute food insecurity among vulnerable populations by further limiting access to essential foods.

In Central Asia, El Niño conditions are typically associated with above-average precipitation amounts between May and September, which generally bodes well for yields, but also increasing the risk of floods and landslides, with ensuing localized crop and livestock losses. At the same time, although the region is not directly reliant on imports from the Gulf, disruptions to the Strait of Hormuz are transmitted through global markets by driving up fuel and fertilizer prices. This, in turn, raises production and transport costs and may reduce fertilizer use. In Kazakhstan, El Niño conditions will likely affect both the planting and growing stages of the major spring wheat crop from May to August. Similarly, in Tajikistan, Uzbekistan, Kyrgyzstan and Turkmenistan, El Niño-related precipitation anomalies between May and September may disrupt key crop growth stages, increasing the likelihood of flooding, with adverse effects on agricultural production and pasture conditions. Disruptions to the Strait of Hormuz could further constrain fertilizer availability and, combined with higher import costs, could reduce fertilizer application, with negative implications for crop yields.

South and Central America

In South America, imports of fertilizer and crude oil via the Strait of Hormuz are generally limited, with the notable exception of Brazil, which sources about 10 percent of its fertilizers and one-fourth of its oil from the region.^{xxiv} Nevertheless, higher global input costs could lead to reduced use of agricultural inputs, particularly in countries with intensive fertilizer use such as Brazil, Colombia and Guyana. Dry weather conditions are forecast across large parts of Brazil,

except the southern region, between September and November, coinciding with the planting period of the 2027 minor season maize crop, which accounts for about 20 percent of the annual production. If agricultural commodity prices remain relatively low, tightening margins could discourage plantings, with the impact to be compounded by soil moisture deficits. In Colombia, below-average precipitation amounts are forecast in northern areas during the third quarter of 2026, with dry weather conditions expected to extend to most of the country in the fourth quarter. These periods coincide with the planting and development stages of the main maize and minor paddy crops, with non-irrigated areas in the northwest likely to be more affected by moisture deficits. In Guyana and the Bolivarian Republic of Venezuela, forecasts of dry weather conditions during the second half of 2026 raise concerns for paddy and maize production, as this period coincides with the entire secondary paddy season in both countries and the main maize season in the Bolivarian Republic of Venezuela. Although paddy cultivation is largely irrigated, prolonged dryness is likely to reduce water availability for irrigation, constraining both plantings and yield potential.

In Central America and the Caribbean, dependence on imports through the Strait of Hormuz is negligible. However, elevated international prices of fertilizers and fuel may constrain input use and farm operations, especially in Guatemala and Honduras, where fertilizer application rates are relatively high. El Niño-associated dry weather conditions are forecast from June to November across Guatemala, El Salvador, Honduras and Nicaragua, coinciding with the critical flowering and grain-filling stages of the main maize crop as well as the entire main bean season, thereby increasing risks to crop yields and production. In the Caribbean, dry weather conditions are also expected between June and November in Haiti and the Dominican Republic, raising concerns for both planting and development of the second minor paddy crop. Concerns are particularly pronounced in Haiti, where irrigation use is very low. Despite not importing fertilizers or fuel through the Strait of Hormuz, fuel prices, and consequently public transport fares, increased by between 25 and 37 percent since early April in response to rising international prices, adding pressure on livelihoods and costs throughout agricultural value chains.

Notes

- i. **NOAA**. 2026. *El Niño/Southern Oscillation (ENSO) diagnostic discussion - ENSO Alert System Status: El Niño Watch*. National Weather Service. Climate Prediction Center/NCEP/NWS. Maryland. 14 May 2026. https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/enso_advisory/ensodisc.shtml
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- iv. **IFPRI**. 2026. *The Iran war's impacts on global fertilizer markets and food production*. International Food Policy Research Institute (IFPRI). Charlotte Hebebrand, Joseph Glauber, Rob Vos, and Brendan Rice. Washington, DC. 1 April 2026. <https://www.ifpri.org/blog/the-iran-wars-impacts-on-global-fertilizer-markets-and-food-production/>
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- vii. **MEI**. 2026. *From Hormuz to the Sahel: A Fertilizer Shock, and a Maghreb Solution*. Middle East Institute (MEI). Jonathan M. Winer. Washington, D.C. 4 May 2026. <https://mei.edu/publication/from-hormuz-to-the-sahel-a-fertilizer-shock-and-a-maghreb-solution/>
- ix. **MEI**. 2026. *The Ripple Effects of the US-Israel War on Iran for North Africa*. Middle East Institute (MEI). Intissar Fakir. Washington, D.C. 23 March 2026. <https://mei.edu/publication/the-ripple-effects-of-the-us-israel-war-on-iran-for-north-africa/>
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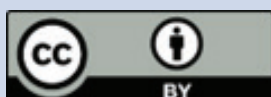
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